

Excel Charts – Visualization Guide

Which chart does what, when to use it, and what it communicates

Goal	Recommended Chart	What it communicates
Compare categories	Column / Bar	Which category is larger or smaller
Show a trend	Line	Increase, decrease, peaks and dips over an ordered axis
Show share of a whole	Pie / Doughnut	Percentage contribution to a meaningful 100% total
Show relationship	Scatter (XY)	Correlation, clusters and outliers between two numeric variables
Show distribution	Histogram / Box & Whisker	Frequency, spread, median, quartiles and outliers
Find major causes	Pareto	Which causes contribute most (80/20 analysis)
Show movement from start to end	Waterfall	How increases/decreases build to a final total
Show process stages	Funnel	Drop-off or conversion through sequential stages
Show hierarchy	Treemap / Sunburst	Contribution across nested categories
Compare different measures	Combo	Two related measures, often Column + Line
Compact row-level trend	Sparklines	Tiny trends inside cells

1. Column Chart

What it does: Compares discrete categories using vertical bars.

Best use: Use for sales by product, marks by student, employees by department. Best when categories are independent.

Avoid / caution: Avoid when there are many categories or when the main story is a continuous time trend.

2. Bar Chart

What it does: Compares categories horizontally.

Best use: Best for rankings, Top 10 lists and long category names.

Avoid / caution: Sort descending when the purpose is ranking.

3. Line Chart

What it does: Shows movement or trend across an ordered axis.

Best use: Best for monthly sales, yearly growth, attendance or website traffic.

Avoid / caution: Do not connect unrelated categories with lines.

4. Pie Chart

What it does: Shows parts of one meaningful whole.

Best use: Use for a small number of shares such as budget or market mix.

Avoid / caution: Avoid many slices; exact comparisons are difficult.

5. Doughnut Chart

What it does: A pie-style part-to-whole chart with a center hole.

Best use: Useful for simple dashboard share visuals.

Avoid / caution: Avoid when several slices are very similar.

6. Area Chart

What it does: Shows trend while emphasizing volume/magnitude.

Best use: Useful for traffic, revenue or cumulative-looking movement over time.

Avoid / caution: Multiple overlapping series can become hard to read.

7. Scatter (XY)

What it does: Plots one numeric variable against another.

Best use: Use for study hours vs marks, price vs sales, height vs weight; useful for correlation and outliers.

Avoid / caution: Both axes should represent numeric variables.

8. Bubble Chart

What it does: Scatter chart with bubble size representing a third numeric measure.

Best use: Example: ad spend vs sales, bubble size = market size.

Avoid / caution: Avoid when bubble sizes are too similar or exact comparison is important.

9. Histogram

What it does: Groups numeric observations into bins to show frequency.

Best use: Use for marks, age, salary or delivery-time distributions.

Avoid / caution: Unlike a column chart, its bins are numerical intervals.

10. Pareto Chart

What it does: Descending bars plus a cumulative percentage line.

Best use: Use for defects, complaints, support issues and 80/20 analysis.

Avoid / caution: Best when you want to identify the vital few causes.

11. Box & Whisker

What it does: Summarizes a distribution using median, quartiles, spread and potential outliers.

Best use: Great for comparing salary/marks distributions across departments or classes.

Avoid / caution: More analytical than presentation-friendly for beginners.

12. Waterfall Chart

What it does: Shows how positive and negative changes transform a starting total into an ending total.

Best use: Excellent for revenue-to-profit, budget changes and financial bridges.

Avoid / caution: Each change should have a clear additive/subtractive meaning.

13. Funnel Chart

What it does: Shows sequential stages of a process.

Best use: Use for Leads → Qualified → Demo → Proposal → Won, or recruitment pipelines.

Avoid / caution: Stages should represent a logical sequence.

14. Radar Chart

What it does: Compares multiple dimensions as a profile.

Best use: Useful for skill profiles or product feature comparisons with a small number of dimensions.

Avoid / caution: Avoid many dimensions or many series; it gets cluttered.

15. Treemap

What it does: Uses nested rectangles to show hierarchy and contribution by area.

Best use: Useful for Company → Department → Product or Region → Category → Sales.

Avoid / caution: Great for hierarchical portfolio/category analysis.

16. Sunburst

What it does: Uses concentric rings to show hierarchy.

Best use: Useful when multiple levels of hierarchy matter.

Avoid / caution: Avoid when exact value comparison is more important than structure.

17. Combo Chart

What it does: Combines chart types, commonly Column + Line.

Best use: Use when comparing related measures such as Sales and Profit Margin %.

Avoid / caution: Secondary axes must be clearly labeled.

18. Stock Chart

What it does: Visualizes financial OHLC/volume data.

Best use: Use for Open, High, Low, Close and volume-style financial data.

Avoid / caution: Not suitable for ordinary category comparisons.

19. Surface Chart

What it does: Shows a response value across combinations of two variables.

Best use: Useful for optimization-style analysis such as price × quantity → profit.

Avoid / caution: Usually unnecessary for normal business reporting.

20. Sparklines

What it does: Tiny charts inside cells.

Best use: Perfect for showing a compact trend beside each employee/product/customer row.

Avoid / caution: Best for quick scanning rather than detailed analysis.

Fast Chart Selection Cheat Sheet

If your question is...	Choose...
Which one is bigger?	Bar / Column
How is it changing over time?	Line
What percentage comes from each category?	Pie / Doughnut
Are X and Y related?	Scatter
How are values distributed?	Histogram / Box & Whisker
Which problems cause most of the impact?	Pareto
How did we get from starting value to ending value?	Waterfall
Where are people/items dropping out of a process?	Funnel
How does a total break down by hierarchy?	Treemap / Sunburst
Do two measures need to be compared together?	Combo
Can I show a mini-trend inside a table?	Sparklines

Common Excel Chart Mistakes

- Using pie charts for too many categories.
- Using a line chart for unrelated categories.
- Overusing 3D effects, which can make comparison harder.
- Adding too many colors, labels or series.
- Using a secondary axis without explaining it.
- Choosing a complicated chart when a simple bar chart would answer the question faster.

Golden rule: Pick the chart based on the question you want the viewer to answer—not on which chart looks fancy.